Advanced Database Management Systems

Unit – 1
Introduction

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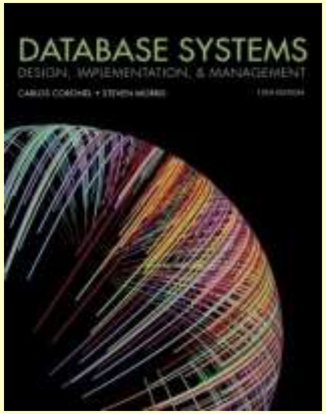
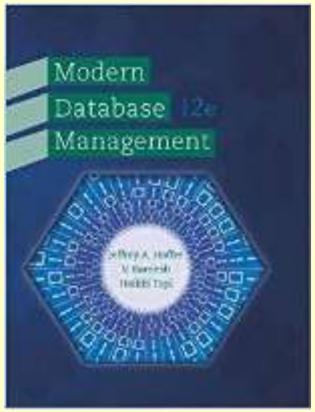
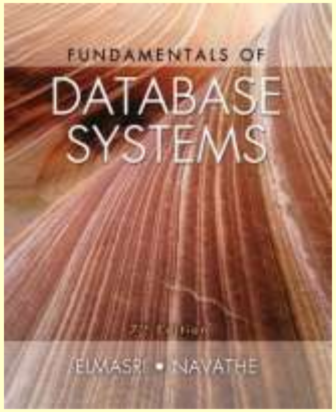
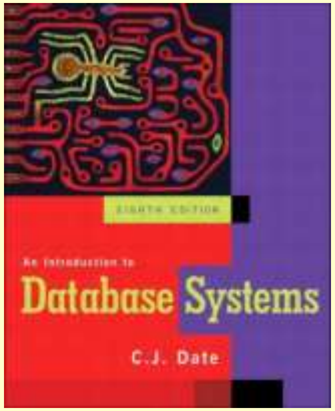
Text & Reference Books

- Database Systems – Design, Implementation and Management: Carlos Coronel, Steven Morris (13th Ed.)
- Modern Database Management: Jeffery. A. Hoffer (12th Ed.)
- Database Systems: Thomas M. Connolly (6th Ed.)
- Fundamentals of Database Systems: Elmasri (7th Ed.)
- An Introduction to Database Systems: C.J. Date (8th Ed.)
- Introduction to PL\SQL: Oracle Press









Course Information

- **Resources:**

  // Google Classroom Code: fx4udn7

- **email: talia.noreen@bnu.edu.pk**

Assessment Category	Marks
Quizzes	20
Assignments	15
Mid term	25
Final Term	40

Course Outline & Schedule

- Some recap lectures where needed
  - SQL querying
- Study of noSQL systems, using MongoDB as an example
- Project presentations

Outline

- Introduction
- Data? Information? Database?
- File Processing System (FPS)
- Drawbacks of FPS
- Database System or Integrated Database Environment(IDE)
- Advantages of IDE
- Types of Databases

Introduction

Dunn's Book of Life (1946)

- “Each person in the world creates a Book of Life. This Book starts with birth and ends with death. Its pages are made up of the records of the principal events in life. Database/Record linkage is the name given to the process of assembling the pages of this Book into a volume.”

Introduction

- **Increasing need of database systems**
- **The Model of Generating/Consuming Data has Changed**

Old Model: Few companies are generating data, all others are consuming data

New Model: all of us are generating data, and all of us are consuming data

FIGURE 1.1 THE PERVASIVE NATURE OF DATABASES

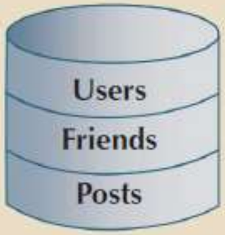
A Day In Susan's Life

See how many databases she interacts with each day

Before leaving for work, Susan checks her Facebook and Twitter accounts

Where is the data about the friends and groups stored?

Where are the "likes" stored and what would they be used for?



On her lunch break, she picks up her prescription at the pharmacy

Where is the pharmacy inventory data stored?

What data about each product will be in the inventory data?

What data is kept about each customer and where is it stored?

After work, Susan goes to the grocery store

Where is the product data stored?

Is the product quantity in stock updated at checkout?

Does she pay with a credit card?

At night, she plans for a trip and buys airline tickets and hotel reservations online

Where does the online travel website get the airline and hotel data from?

What customer data would be kept by the website?

Where would the customer data be stored?

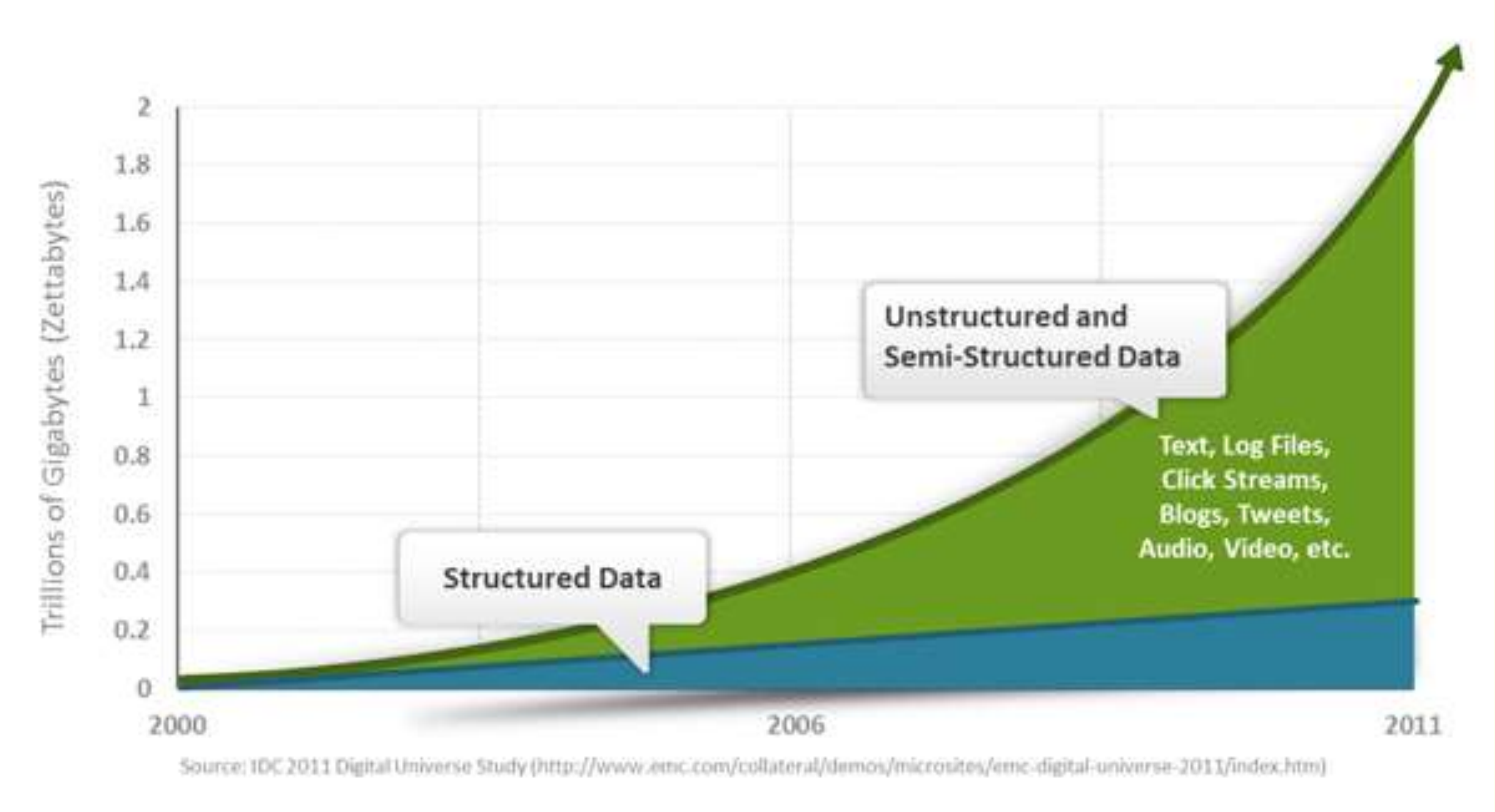
Then she makes a few online purchases

Where are the product and stock data stored?

Where does the system get the data to generate product "recommendations" to the customer?

Where would credit card information be stored?

Exponential Data Growth



Data

- Data refers to the raw facts & figures concerning:
  - ✓ PEOPLE
  - ✓ EVENT
  - ✓ ORGANISATION
  - ✓ OBJECT
- The amount of available data in the modern computer world is literally exploding (3 Vs: Volume, Velocity, Variety)
- The modern data sources/types include multimedia data, sensor’s data, data on web and so on.
- To get most out of data, we require tools than can simplify the task of managing the data and extract useful information out of it in timely fashion.

Data

- Large volume of facts, difficult to interpret or make decisions based on

Class Roster

Course: MGT 500 Business Policy

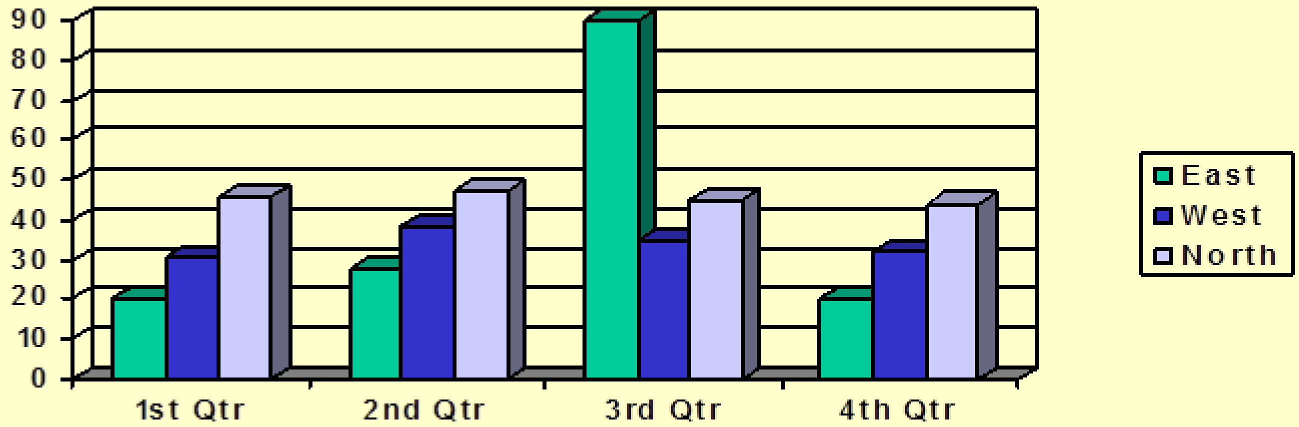
Semester: Spring 200X

Section: 2

Name	ID	Major	GPA
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Data vs Information

- Information is derived from data OR Information is the processed data
- Information may be presented either Textually or Graphically or both



- The representation and amount of information depends upon the requirements of a user.
- When data is processed and organized into a form needed for its application, it is called an Information Product (IP).

Data vs Information

- Accurate, reliable, relevant and timely information is key to good decision making.
- Information is now increasingly viewed as an organizational asset (Redman, 2008) which can be used to improve organizational performance and help an organization gain a competitive advantage in the marketplace.

Data Hierarchy

Field (Data Item)

↓

Record

↓

File

↓

Database

Database

- Database System can simply be regarded as a computerized record keeping system.
- Database System is a system whose overall purpose is to maintain data and make the information available to its users as per their requirements.
- The user of the system can perform the operations like:
  - Defining structure and types of data
  - Data manipulation (Insert, Delete, Update, Query etc.)
  - Enforcing security restrictions
- A Database Management System (DBMS) is a collection of programs that manages the database structure and controls access to the data stored in the database.

Database

- Database is a shared collection of logically related data, and a description of this data, designed to meet the information requirements of multiple users in an organization.
- Database is an organized collection of **data** (and **metadata**) about **entities** and the **relationships** among these entities.
- Metadata: Data that describes properties or characteristics of other data OR it is “data about data”.
- The description of data is also known as system catalog or data dictionary.
- Metadata allows database designers & users to understand what data exists and what the data mean?
- Metadata describes the domain (set of possible values) and the constraints for a data item.

Metadata Example

Data Item		Value			
Name	Type	Length	Min	Max	Description
Course	Alphanumeric	30			Course ID & Name
Section	Integer	1	1	9	Section Number
Semester	Alphanumeric	10	1	8	Semester & Year
Name	Alphanumeric	30			Student name
ID	Integer	9			Student ID
Major	Alphanumeric	4			Student Major
GPA	Decimal	3	0.0	4.0	Student GPA

Examples of Database Applications

- Databases play a critical role in almost all areas where computers are used.
- Databases are everywhere …
  - Student / Employee's Information System
  - Library System
  - Hotel / Airline Reservation System
  - Billing System
  - Stock / Inventory System
  - Payroll System
  - Geographical Information Systems (GISs)
  - Data Warehouses

Record Keeping Techniques

1. Manual Record Keeping.
2. Computerized Record Keeping
   a) File Processing System.
   b) Database System or Integrated Database Environment. (The paradigm shift)

Case Study

- Consider a College System consisting of the following offices:
  - Admission Office
  - Academics Office
  - Exam Office
- Each Office is maintaining its own set of files for its day to day operations.
- Some of the files used in the system are Student's File, Faculty File, Course File, Correspondence File etc.

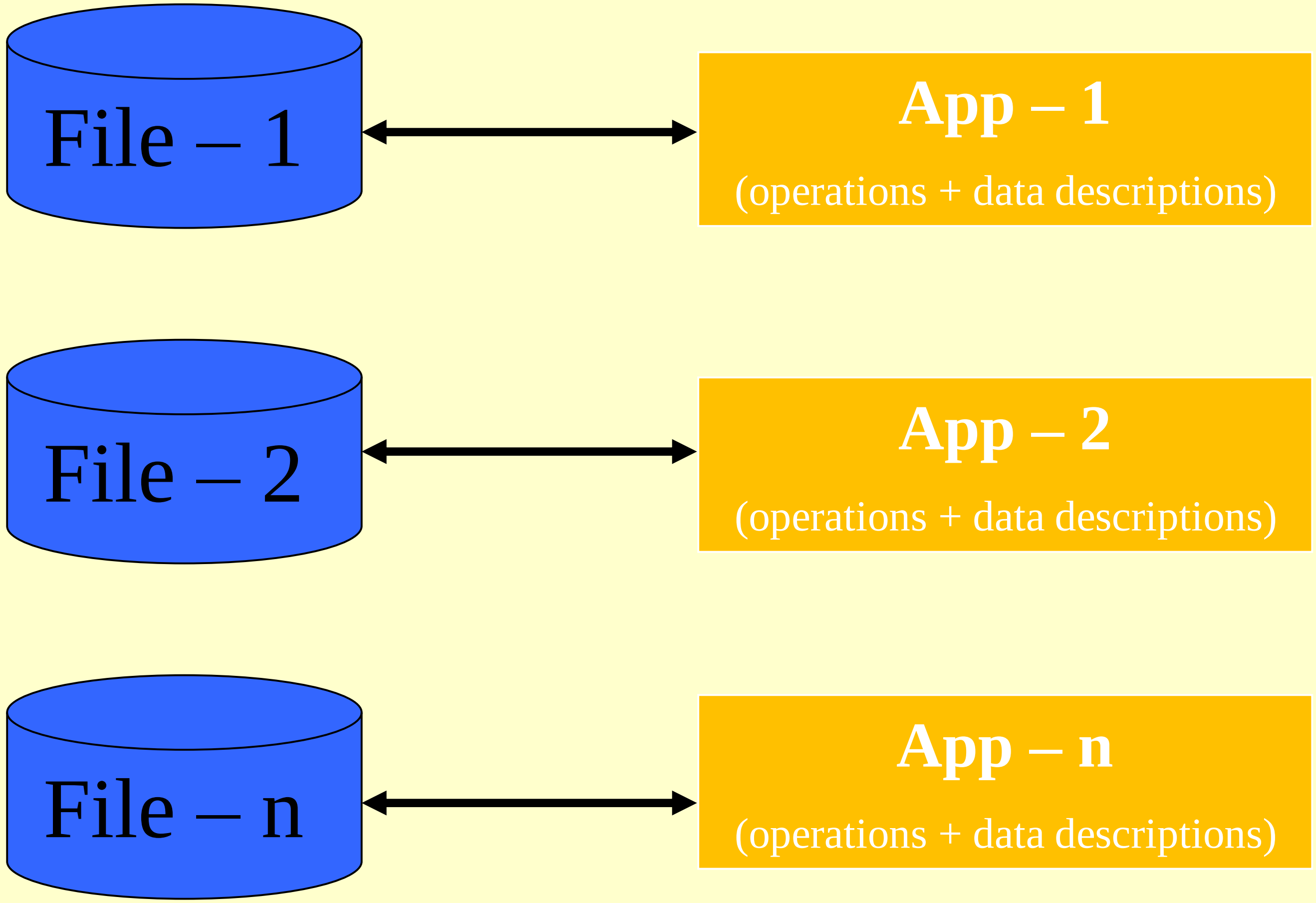
Manual System - Drawbacks

- High data volume
- Not reliable
- Inefficient
- Duplication of data
- Inconsistency
- A lot of data movement is required
- The System can’t answer complex queries involving multiple departments.

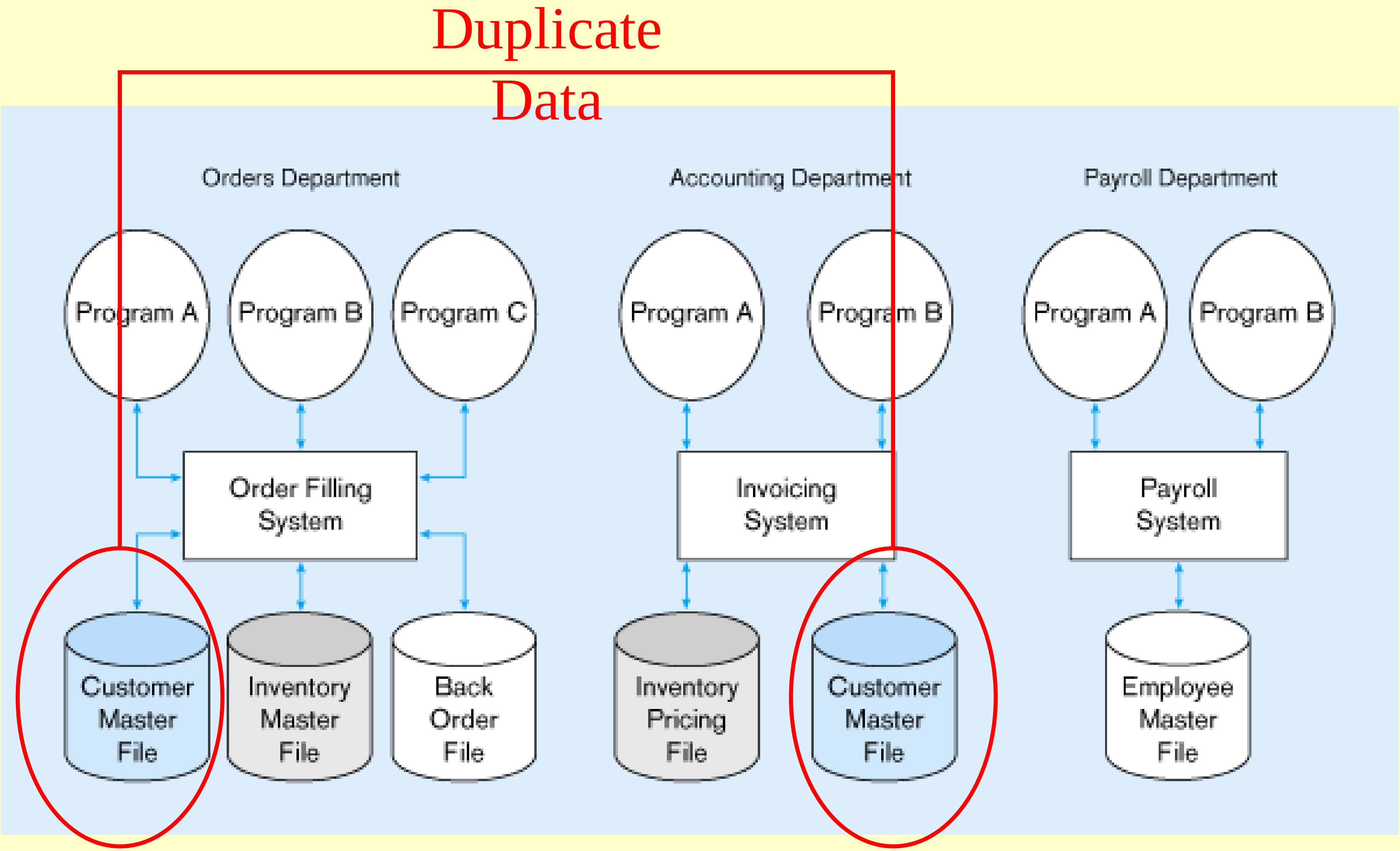
File Processing System

- File-based systems were an early attempt to computerize the manual filing system.
- The manual files were replaced by computer files.
- A person responsible for developing and managing computerized file processing system was called Data Processing (DP) specialist.
- In the traditional file processing system, the existing manual system is automated by focusing on the data processing needs of the individual departments instead of treating the organization as whole.
- Each application will have its own set of Private Files designed to meet the needs of a particular department.

File Processing System



File Processing System



Drawbacks of File System

1. Redundancy of Data
2. Inconsistent Data
3. Poor Enforcement of Standards
4. Data Dependence
5. Limited Data Sharing
6. Extensive Development & Maintenance Effort
7. Limited Support for Queries
8. Inflexible
9. Lack of provision for security
10. Limited recovery from failure

Drawbacks of File System

1. Redundancy of Data

- Redundancy means duplication of data.
- Since applications in FPS are independent so unplanned duplicate data is there.
- Same information is needed to be kept at different places. It costs time and money.
- Same data may have to be input several times to update all the occurrences.
- Redundancy results in data anomalies (Insert, Update, Delete)

Drawbacks of File System

2. Data Inconsistencies or Data Anomalies

- Redundancy leads to data inconsistency or data anomalies.
- A data anomaly occurs if an operation (update, insert, delete) has not yet been performed against all the occurrences. Consequently, same data stored at different places will disagree with each other
- The use of different programming environment and platforms lead to Heterogeneous Environment.

3. Poor Enforcement of Standards

- Organization wide enforcement of standards is poor. This leads to the following types of inconsistencies:
  a) Synonym: Using different names for same data item
     Example: Stu-Id, Reg-No.
  b) Homonym: Using same name for different data items.
     Example: “course” for a single subject or entire program.

Drawbacks of File System

4. Data Dependence

- The definition of data is embedded in the application programs, rather than being stored separately and independently.
- The applications are constrained to work only with the given file description. Any change in the file structure or data requires changes to all the applications using that file. Such applications are called Data Dependent Applications.
- Data dependence leads to excessive program maintenance.
- Often difficult to locate all programs affected by change
- Errors are often introduced when making changes.
- Incompatible file structures (use of different programming languages) makes the situation even worse.

Drawbacks of File System

5. Limited Data Sharing

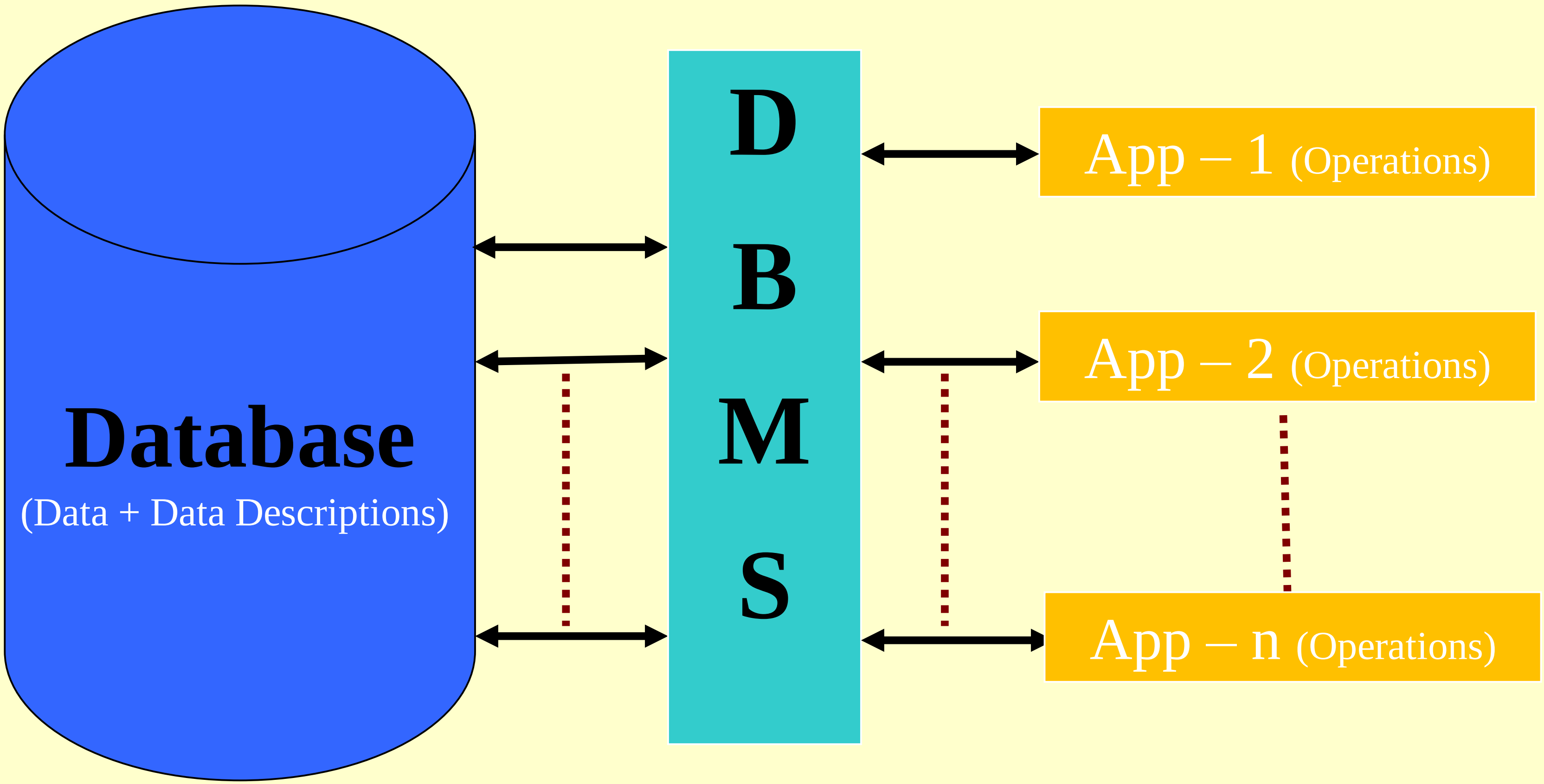
- As each application has its own private files so little opportunity to share data with others.
- The filing system works well when we have to interact with each file separately. However, it breaks down when we have to cross-reference or process the information in the files. This makes it hard to answer queries involving multiple files.

6. Extensive Development & Maintenance Effort
7. Limited Support for Queries
8. Inflexible
9. Lack of provision for security
10. Limited recovery from failure

Integrated Database Environment(IDE)

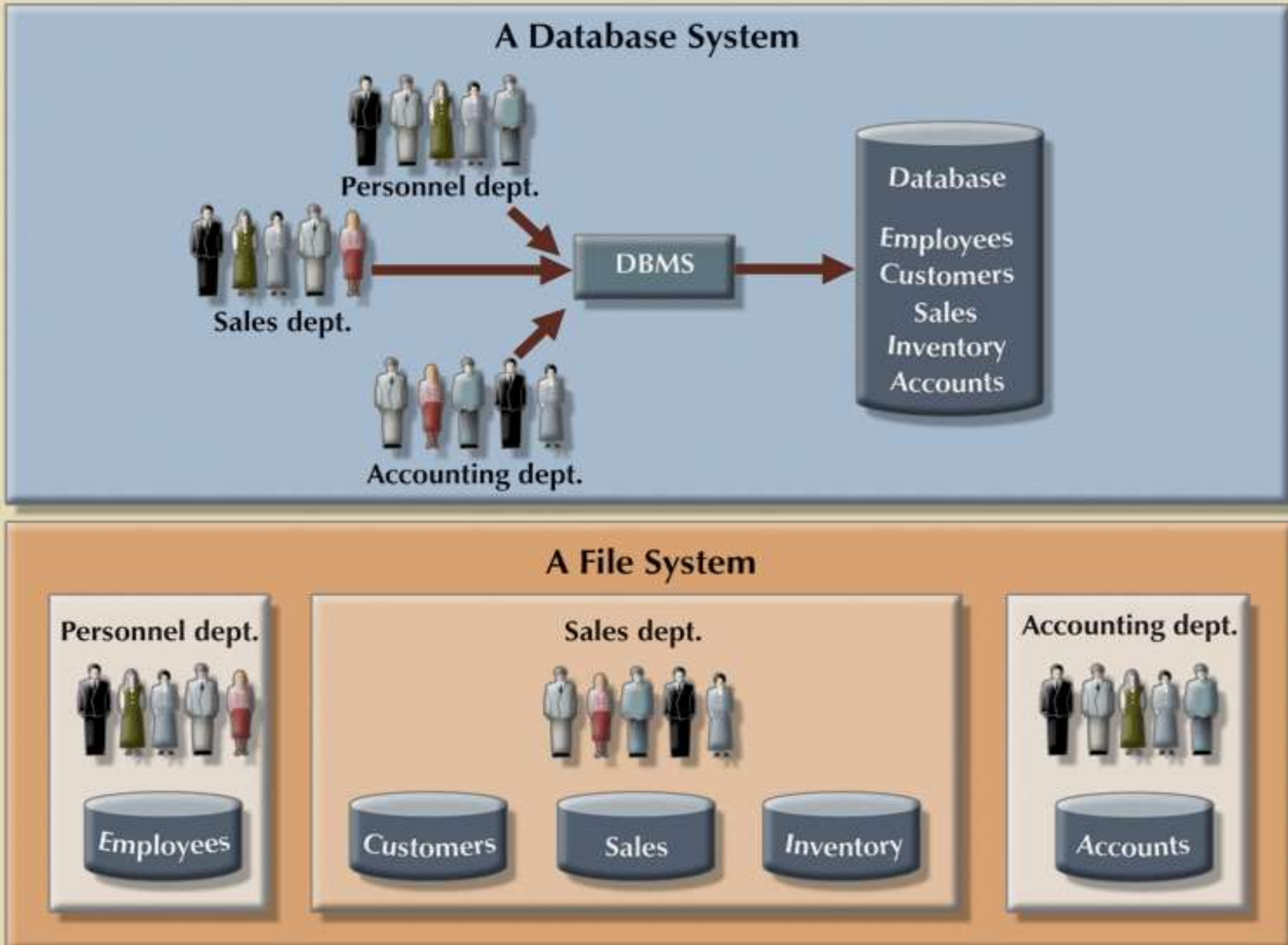
- The limitations of file-based approach can be attributed to the following factors:
  - Redundancy of data
  - Data dependence
- In contrast to file processing system, the IDE has a single large repository of data, where data definition is separated from application programs.
- Emphasizes the integration & sharing of data throughout the organization.
- The organization wide requirements are analyzed as a whole and there is no longer concept of MY FILE or Private Files.
- A Database Management System (DBMS) is a software designed to assist in maintaining and utilizing large collections of data.
- By storing data in a DBMS rather than as a collection of files, we achieve enormous advantages.

Database System



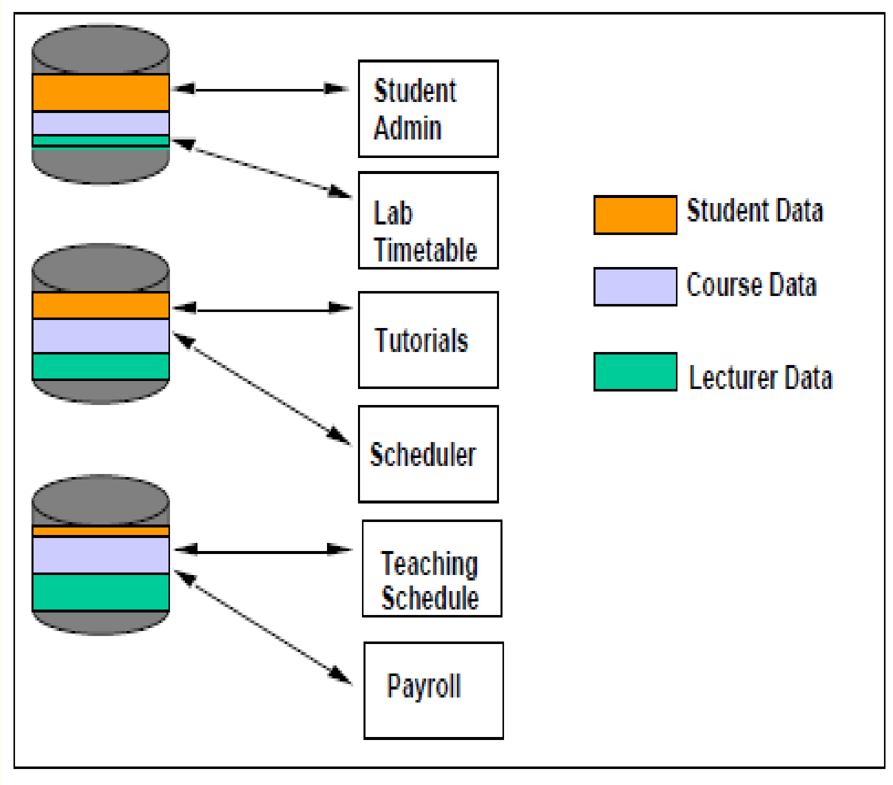
DBMS manages data resources like an operating system manages hardware resources

FIGURE 1.6 Contrasting database and file systems

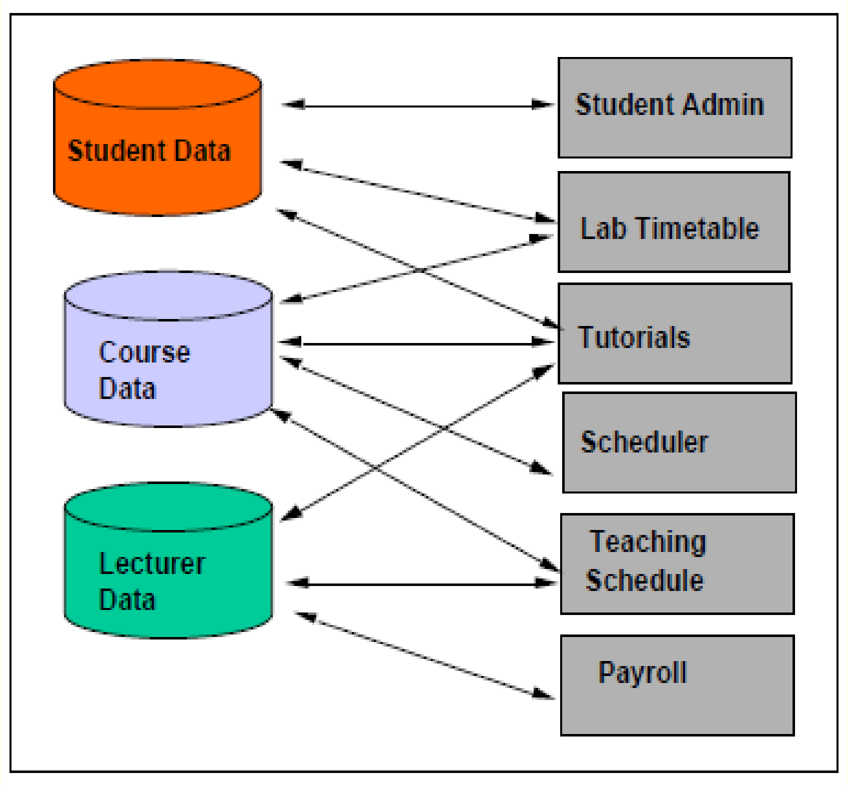


Contrasting Database and File System

File System



Database System



Adapted from Lecture slides of Werner Nutt

Advantages of Database Approach

1. Reduced or Controlled Data Redundancy
2. Improved Data Consistency
3. Enforcement of Standards
4. Reduced Program Maintenance
5. Data Sharing
6. Data Integrity (Improved Data Quality)
7. Improved Security Restrictions
8. Improved Accessibility & Responsiveness
9. Improved Decision Making
10. Efficient Query Processing
11. Backup & Recovery
12. Data Independence

Advantages of Database Approach

1. Reduced or Controlled Data Redundancy

- Since all the data items are stored in a single database, therefore Redundancy is minimized.
- Separate data files are integrated into a single, logical structure to reduce redundancy
- 100% elimination of redundancy is not possible in order keep the logical connections among the data items.

2. Improved Data Consistency

- Reduction in redundancy automatically avoids inconsistency.
- Wang and Wand describes data consistency as “a data value can only be expected to be same for the same situation”.
- If a data item appears only once, any change to its value needs to be performed only for once and the database will always be in some consistent / correct state.

Advantages of Database Approach

3. Enforcement of Standards

- This is possible as database is designed to meet the organization wide requirements.
- The standards can be name of data items and their format, data codes, documentation standards, operation standards, security policies etc.

4. Reduced Program Maintenance & Development Cost

- When all the organization's requirements are satisfied by one database instead of many separate files, the maintenance and development cost automatically reduces.

Advantages of Database Approach

5. Data Sharing

- Sharing means that the same data source is used by multiple applications.
- Data is centralized and hence can be shared not only by the existing applications but also new applications can be developed to operate against the same data.
- The applications that can access same data simultaneously are called Online Transaction Processing (OLTP) applications and DBMS uses Concurrency Control Mechanism to ensure that multiple users can access and update data correctly.

Advantages of Database Approach

6. Data Integrity (Improved Data Quality)

- Data integrity refers to the correctness of data.
- Integrity constraints or rules ensure that the data stored in the database is purified and accurate.

7. Improved Security Restrictions

- Database security is the protection of database from unauthorized disclosure, alteration or destruction.
- DBMS provides strong security measures against such threats. Some of them are:
  - Password checks
  - User defined procedures
  - Defining user privileges
  - Audit trial system
  - Data encryption

Advantages of Database Approach

8. Improved Accessibility & Responsiveness.

- In File Processing System, data accessibility is quite difficult as it is procedural based. You should know the detailed procedural steps. HOW TO DO?
- On the other hand, accessing data is lot easier in DBMS with the help of a non procedural language – SQL. You only need to know the simple commands or in other words WHAT TO DO?

9. Improved Decision Making

- Consolidated reports
- Ad-hoc queries
- Multiple views of data

Advantages of Database Approach

10. Efficient Query Processing

- Specialized data structures and search techniques
- Indexing
- Buffering or caching module
- Query processing and optimization

11. Backup & Recovery

- The backup and recovery subsystem of DBMS provides efficient techniques for database recovery.

Advantages of Database Approach

12. Data Independence.

- The separation of data descriptions from the applications using the data is called data independence.
- Data Independence can also be defined as: "The immunity of an application to change in storage structure – which implies that the applications don't depend on any particular storage structure".
- Allows change & evolution of database systems without changing the application programs
- Without data independence, it is must to understand the arrangements of data elements in order to perform an operation against them.
- Different data arrangements will need different algorithms even for the same operation.

Advantages of Database Approach

12. Data Independence.

- Clearly, data dependence is not desired because of the following reasons:

a) Each user should be able to access data in a customized way.

b) Users should not have to deal directly with the physical database storage details.

c) Different application may need different physical structure for the efficiency of their operations.

d) The DBA should have freedom to change the storage structure or access technique or both in response to changing requirements without much disturbing the existing applications.

Cost & Risk of Database Approach

1. Need for new specialized personnel
2. Cost of DBMS and additional hardware cost
3. Installation cost & complexity
4. Conversion from legacy system to modern database system
5. Organizational conflict
6. Higher impact of failure
7. Not suitable for applications with tight real-time constraints

Types of Databases

- Databases can be categories on the basis of the following criterions:

Criteria	Types
1. Number of users	• Single User Database • Multi-user Database
2. Data location	• Centralized Database • Distributed Database
3. Data usage	• Operational (OLTP) • Analytical (OLAP)
4. Form of data	• Unstructured • Structured • Semi-structured (XML)

Types of Databases

Single user with desktop computer → Mainframe computer with thousands of users

- Categories are:
  - Personal computer databases(Single User)
  - Workgroup databases(Multi-user)
  - Department databases (Multi-user)
  - Enterprise database (Multi-user)

Types of Databases

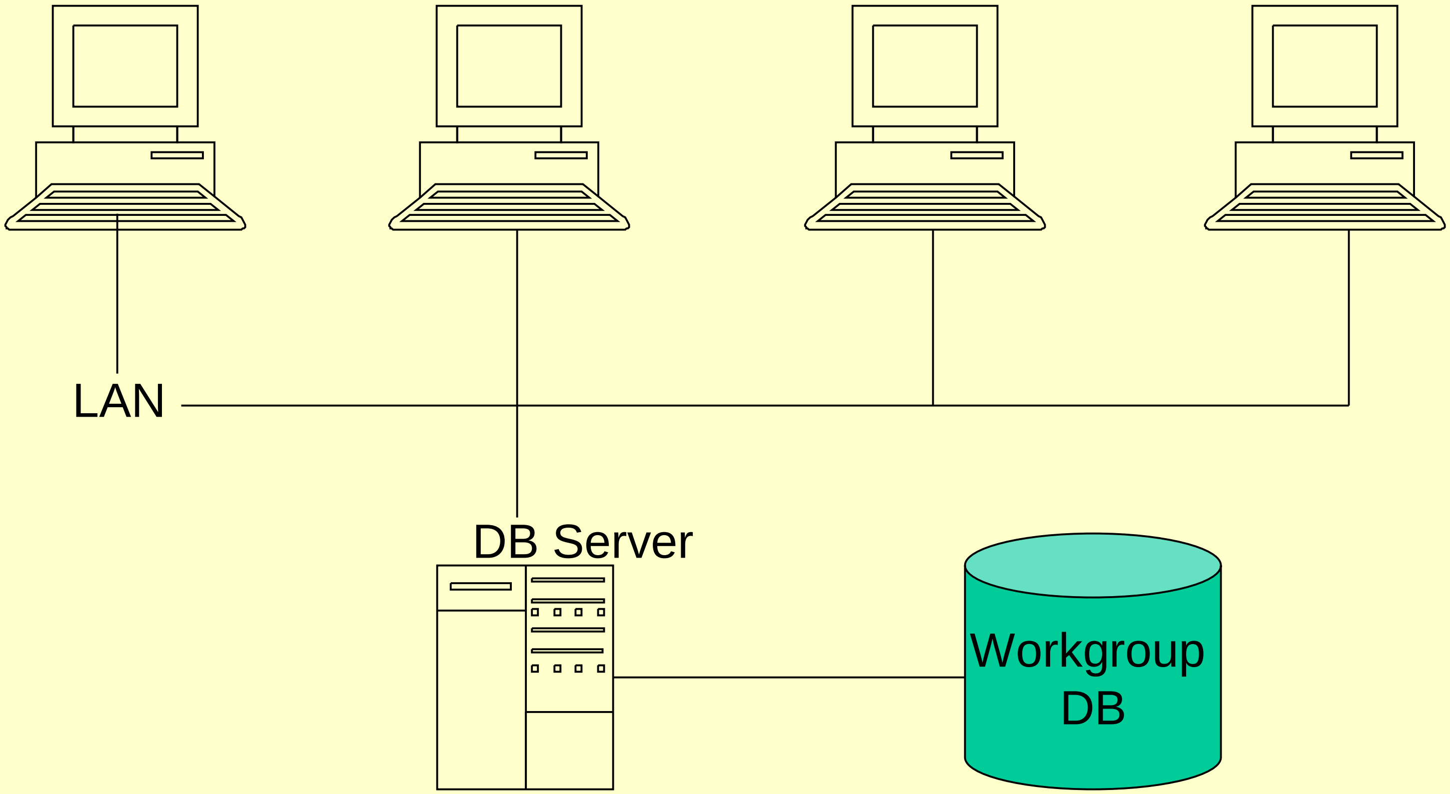
Personal Computer Databases

- Support one user with a standalone PC
- E.g. a student's own database or a sales person's simple database

Workgroup Databases

- Workgroup: relatively small group of people who collaborate on same project/application.
- A workgroup DB supports the collaborative efforts of a workgroup.
- Allows data sharing.
- Its model is shown on the next fig:

Continued…



Workgroup DB on LAN

(Method of Data Sharing)

Types of Databases

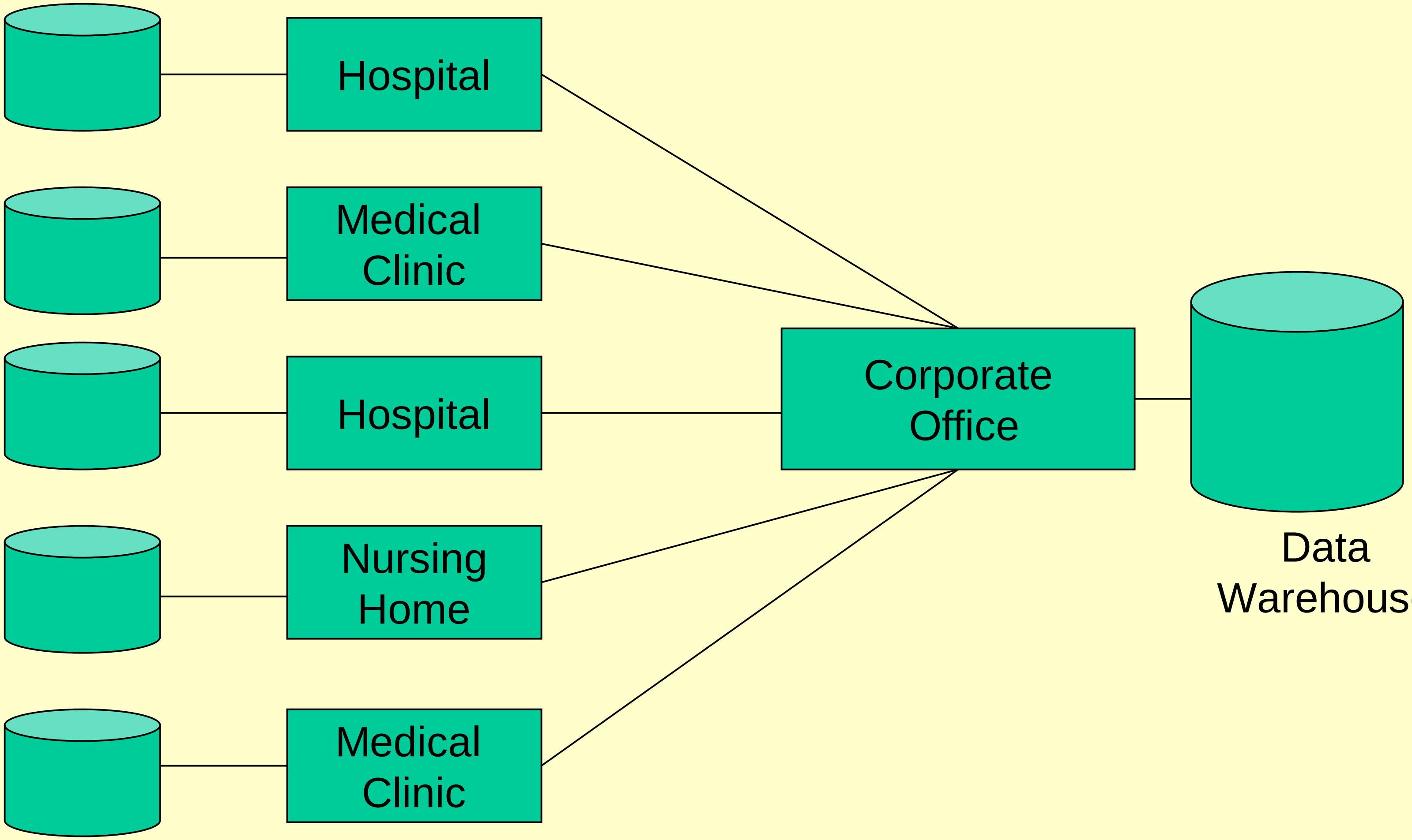
Department Databases

- Department: Functional unit within an organization
- Department DBs support function & activities of a department.
- E.g. personnel DB to track employees, jobs, skills etc.

Enterprise Databases

- DB scope is the entire organization
- To support organization-wide operations & decision making.

Continued…


Hospital
Medical Clinic
Hospital
Nursing Home
Medical Clinic
Corporate Office
Data Warehouse

Types of Databases

Internet, Intranet & Extranet Databases

- Internet: A worldwide network that connects users of multiple platforms easily through an interface known as web browser
- Extranet: Use of Internet protocols to establish limited access to company data & information by the company’s customers & suppliers
- Intranet: Use of internet protocol to establish access to company data and information that is limited to the organization

Centralized & Distributed Databases

- **Centralized** database system – supports a database located at a single site
- **Distributed** database system – supports a database distributed across several different sites

Types of Databases

- The past few years have produced advances in technology leading to exciting new applications of database systems such as:
  - Multimedia databases
  - Geographic information systems
  - Data warehouses
- Multimedia databases – store pictures, video clips, and sound messages
- Geographic information systems (GIS) – store and analyze maps, weather data, and satellite images
- Data warehouses & on-line analytical processing (OLAP) is used to extract and analyze useful information from very large databases for decision-making.

Reading Material

- Text Book: Chapter 1 (1.1 to 1.6)
- Ref. Book-A: Chapter 1 (page 2 to page 15)
- Ref. Book-B: Chapter 1 (1.1 to 1.6)
- Ref. Book-C: Chapter 1 (1.1 to 1.3, 1.6)